

PLSC 197 / SODA 110N: Social Data, Technology, and Artificial Intelligence

Applied Lab One: Data Visualization Example

I received a request for a “template” – an example of what a completed Lab One assignment might look like. So I decided to make one.

To do this, I’m going to use a different dataset: The 2024 [American National Election Study](#). This is a semi-annual survey of a national probability sample of adults in the U.S. In this case, the data have about 1700 observations (survey respondents).

Here are the variables I chose:

- Discrete: **Sex**: The respondent’s self-identified sex; listed as “Female” or “Male.”
- Continuous: **Age**: The respondent’s age in years. Ranges from 18 to 94.
- Continuous: **LaborTherm**: The respondent’s “[feeling thermometer](#)” score towards “labor unions.” The values range from 0 to 100.

A couple notes about this example:

1. The fact that you might not know what these variables are isn’t really that important; the point is to describe the “shape” of the data, so these examples and descriptions can be useful.
2. I cannot emphasize this enough: **Please do not think of this template as a “recipe” for how to complete this assignment.** There are literally an infinite number of ways to complete the lab well; don’t think that you need to (or even should) mimic this example down to the last detail. That means:
 - You aren’t *required* to use pie charts, boxplots, and a scatterplot (though you are welcome to do so);
 - The plots elements don’t need to be green
 - Your font labels don’t need to be in Arial font
 - You do not have to use Claude
 - You don’t need to separate your prompts with ---
 - etc.

The larger point is, use this, but don’t really think of it as a “template.” It’s here as a guide, but please do not try to emulate it pixel by pixel.

The example begins on the next page.

SODA 110N: Social Data, Technology, and Artificial Intelligence

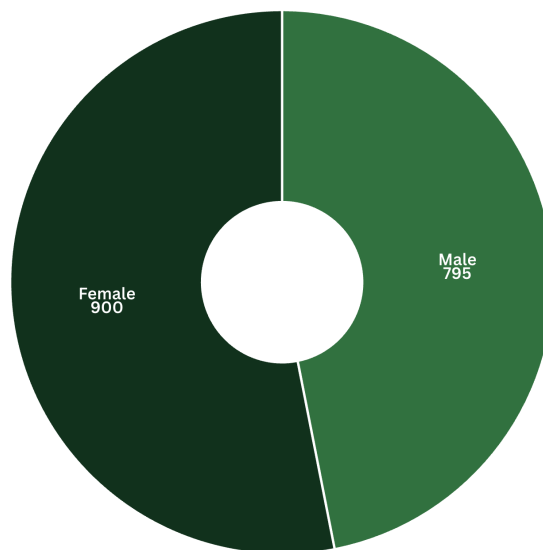
Applied Lab One: Data Visualization

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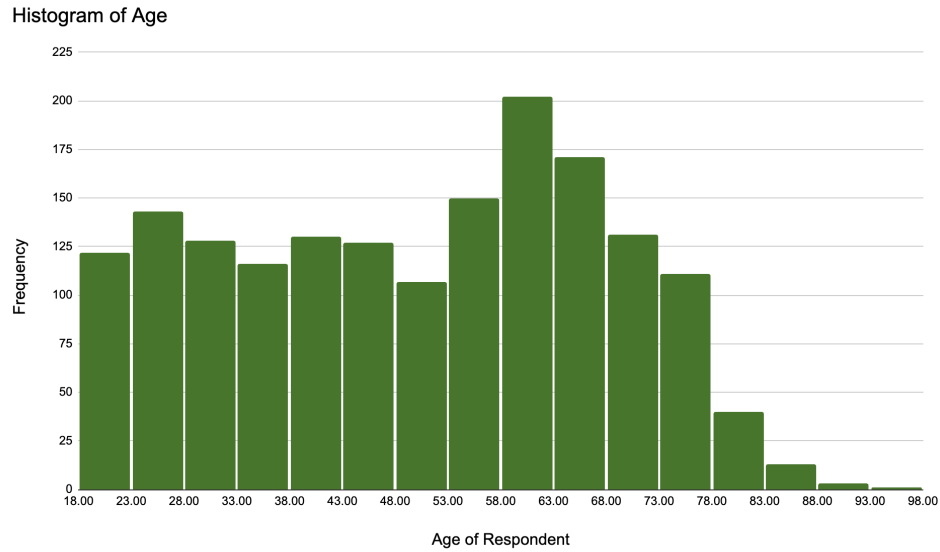
Applied Lab:

1. For this assignment, the variables I chose were *Sex* (discrete variable, D), *Age* (continuous variable, X), and *Labor Union Feeling Thermometer* (continuous variable, Y).
2. A barplot of *Sex* is presented below. The plot was made using Flourish.



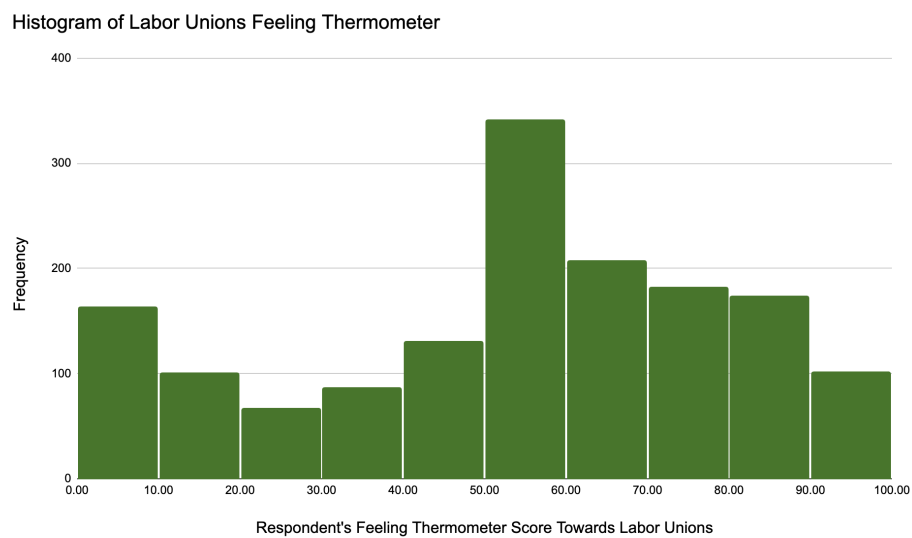
The plot indicates that 900 out of 1695 respondents (53.1 percent) in the survey were female, and 795 (46.9 percent) were male.

Next, I present a histogram of the distribution of *Age*. The plot was made using Google Sheets.



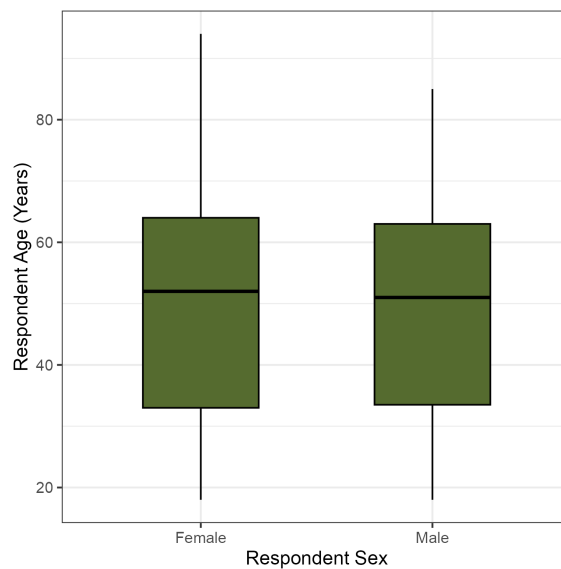
The histogram indicates that there are roughly equal numbers of respondents in the age ranges 18-23 through 48-53; there are slightly more respondents in the ranges from 53-58 through 63-68, above which the numbers of respondents decrease relatively rapidly.

Next is a similar histogram of the distribution of *Labor Union Feeling Thermometer*. This plot was also made using Google Sheets.



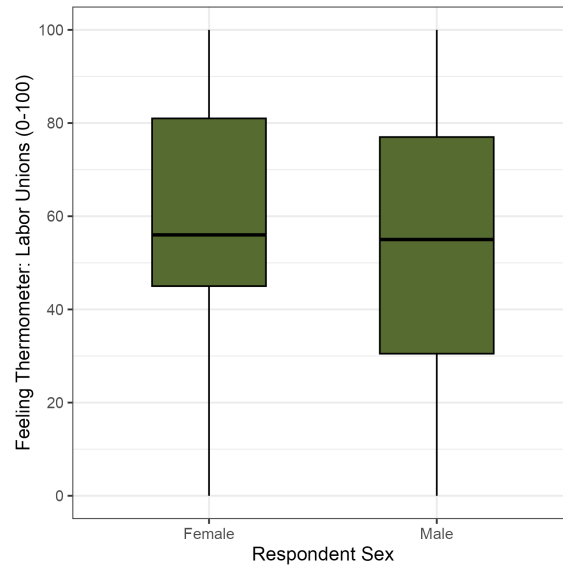
The plot suggests a relatively small number of respondents have thermometer scores for organized labor in the 10-40 range. There is a small uptick at the bottom of the scale (0-10), and a substantial number of respondents giving scores in the 50-90 range, with the single largest category being those giving scores in the range of 50-60. The numbers tail off at the highest values.

3. For the bivariate plots, I start with plotting *Age* conditional on *Sex*. I do this via conditioned boxplots, generated using Claude Opus 5.5. The text of the prompts used to generate the plots in this part of the lab are reproduced at the end of the assignment.



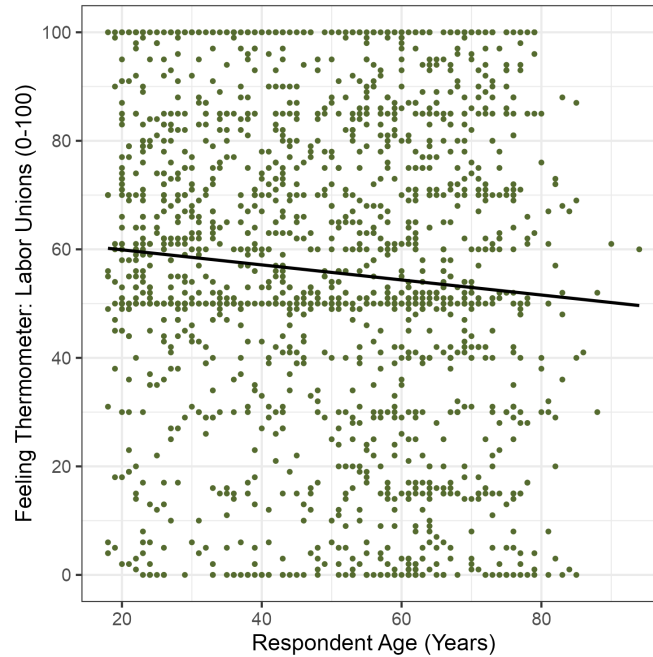
The boxplots show that the median age of respondents in the survey is almost exactly the same for women and men. Male respondents have a slightly narrower range of values of *Age* than women.

Next is a similarly conditioned boxplot of *Labor Union Feeling Thermometer* conditioned on *Sex*.



Once again, the median values of *Labor Union Feeling Thermometer* are almost identical across the two groups. However, female respondents' scores are on average higher than males', and we see relatively few women respondents who have low (below 40) thermometer scores toward unions. Male respondents' scores are more symmetrically distributed around their median.

The last plot is a scatterplot of *Labor Union Feeling Thermometer* (vertical axis) plotted against *Age* (horizontal axis).



This plot does not show a clear pattern; across much of the age range, there is a similar pattern of scores on *Labor Union Feeling Thermometer*. The exception is at higher ages, where *Labor Union Feeling Thermometer* shows a decrease on average. This is reflected in the black trend line shown in the plot; on balance, older people in the survey are very slightly less likely to have high thermometer scores towards unions than are younger people.

Prompts

The AI-created part of this assignment was generated using Claude Opus 5.5. The prompts (separated by ---) used were as follows:

I am going to need to create some visualizations from the attached data, which come from the 2024 American National Election Study pilot data. The three variables are pretty self-explanatory: respondent sex, respondent age in years, and respondent's 0-100 "feeling thermometer" score toward labor unions. I will give specific instructions about the visualizations; they are for a class assignment. In the meantime, please do nothing until instructed.

The first visualization should be a conditioned boxplot. The variable to plot is Age, and the variable to condition on is Sex. So, plot side-by-side boxplots of Age, by Sex. Make the plot fill (boxes, outliers) a dark olive green. Label the axes appropriately, in Arial font.

Thank you. Next, please make a conditioned boxplot where the variable to plot is LaborTherm, and the variable to condition on is Sex. So, once again plot side-by-side boxplots of LaborTherm, by Sex. Make the plot fill (boxes, outliers) a dark olive green. Label the axes appropriately, in Arial font. You need not create any R code; just draw the plot, please.

Finally: Create a scatterplot of Age by LaborTherm. Color the (small) points similarly to the last two plots, and label the axes appropriately. Also, include a black line that shows the association between the two variables.

Please place LaborTherm on the vertical axis, and age on the horizontal axis. Otherwise, leave the plot the same.